

sHoTT algebra

mark

May 12, 2025

1 Modal Type Theory

1.1 Motivation

For building an internal universe in type theory, classifying fibrations for instance, we would love to have a right adjoint to exponentiation by an interval $(\mathbb{I} \rightarrow _)$.

But this is not possible, and can be proven internally to type theory. This is because the right adjoint isn't stable under substitution (pullback) which all type theoretic constructs are.

One way to solve this is to have a modality; an operation not stable under substitution. E.g. we could have $\Box$ which takes a type (interpreted as a presheaf) to its constant presheaf of global points. Then we could state internally that

$$\Box(\mathbb{I} \rightarrow Y) \rightarrow X \simeq \Box Y \rightarrow (X_{\mathbb{I}})$$

To introduce such a $\Box$ operation we must change the type theory.

2 Subcategories

Theorem 2.1 *Let $C :_{\flat} \mathcal{U}$ be a category. and $\phi :_{\sharp} \langle \flat \mid C \rangle \rightarrow \text{Prop}_{\square}$. Define*

$$C_{\phi} = \sum_{c:C} \langle \sharp \mid \phi(c^{\eta}) \rangle$$

Then if $a, b : C_{\phi}$ we have $\text{hom}_{C_{\phi}}(a, b) = \text{hom}_C(\pi_1(a), \pi_1(b))$.

Proof. We have

$$\begin{aligned} \mathbb{I} \rightarrow \sum_{c:C} \langle \sharp \mid \phi(c^{\eta}) \rangle &= (f : \mathbb{I} \rightarrow C) \times ((i : \mathbb{I}) \rightarrow \langle \sharp \mid \phi(f(i)^{\eta}) \rangle) \\ &= (f : \mathbb{I} \rightarrow C) \times \langle \sharp \mid (i : \langle \flat \mid \mathbb{I}^{\eta} \rangle) \rightarrow \phi(f(i)) \rangle \\ &= (f : \mathbb{I} \rightarrow C) \times \langle \sharp \mid (b : 2) \rightarrow \phi(f(b)) \rangle \\ &= (f : \mathbb{I} \rightarrow C) \times \langle \sharp \mid \phi(f(0)) \times \phi(f(1)) \rangle \\ &= (f : \mathbb{I} \rightarrow C) \times \langle \sharp \mid \phi(f(0)) \rangle \times \langle \sharp \mid \phi(f(1)) \rangle \end{aligned}$$

That is morphisms in C_{ϕ} are morphisms of C whose endpoints are in C_{ϕ} . □

3 Reflective Localisations

Definition 3.1 We say a map $f : C \rightarrow D$ is a localisation if it admits a fully faithful right adjoint $g : D \rightarrow C$.

Lemma 3.2 *The following are equivalent:*

- $f : C \rightarrow D$ is a localisation.
- There exists $g : D \rightarrow C$ and a natural isomorphism $\varepsilon : f \circ g \rightarrow \text{id}_D$
- f has a right adjoint $g : D \rightarrow C$ so that $f \circ g : D \rightarrow D$ is an equivalence.

Proof. (1 $\Rightarrow$ 2) Suppose $f : C \rightarrow D$ is a localisation. Then we must show that the unit of the adjunction $\varepsilon : f \circ g \rightarrow \text{id}_D$ is an equivalence.

Proposition 3.3 *If $f : C \rightarrow D$ is a localisation, then for any category E , the pullback $f^* : E^D \rightarrow E^C$ is fully faithful.*

Proof. We have a chain of equivalences

$$\text{Hom}(a \circ f, b \circ f) \simeq \text{Hom}(a \circ f \circ g, b \circ f \circ g) \simeq \text{Hom}(a, b)$$

as $f \circ g$ is an equivalence. □

4 Symmetric monoidal categories

Definition 4.1 The category of finite types is given by

$$\mathcal{F} := \mathcal{S}_\phi = \sum_{X:S} \langle \# \mid \phi(X^\eta) \rangle$$

where

$$\phi(X) = \boxtimes \exists_{n:\mathbb{N}} X = \text{Fin}(n)$$

We can similarly define $\mathcal{F}_*$ the category of pointed finite types.

Definition 4.2 A morphism $f : (X, x_0) \rightarrow (Y, y_0)$ in $\mathcal{F}_*$ is inert if

$$\prod_{y:Y} (y \neq y_0 \rightarrow \text{isContr}(\text{fib}_f(y)))$$

Definition 4.3 For each $(X, x_0) : \mathcal{F}$ and $x : X$ with $x \neq x_0$ we have a pointed map $\delta_x : (X, x_0) \rightarrow \langle 1 \rangle$ given by sending x to 1 and everything else to 0. A symmetric monoidal category is a cocartesian fibration $\mathcal{C}^\otimes : \mathcal{F}_* \rightarrow \mathcal{U}$ satisfying: the canonical map

$$\mathcal{C}^\otimes(X, x_0) \xrightarrow{\langle \delta_x \rangle_x} \prod_{x:X \setminus x_0} \mathcal{C}^\otimes(\langle 1 \rangle)$$

is an equivalence

Example 4.4 Let $\mathcal{C}$ be a category with finite products. Then $\mathcal{C}$ has a natural symmetric monoidal structure

Proof. Define a family;

$$\begin{aligned}\mathcal{C}^\times : \mathcal{F}_* &\rightarrow \mathcal{U} \\ \mathcal{C}^\times(X, x_0) &= \mathcal{C}^{X \setminus x_0}\end{aligned}$$

This is a cocartesian fibration. Let $X, Y : \mathcal{F}_*$, $u : \text{hom}_{\mathcal{F}_*}(X, Y)$, $e : \mathcal{C}^{X \setminus x_0}$. By directed univalence, $u : X \rightarrow Y$ is a function. We then define an object of $\mathcal{C}^{Y \setminus y_0}$ by, for $y : Y \setminus y_0$, we have $e'_y := \prod_{x : \text{fib}_u(y)} e_x$ where $\prod$ here is the product in the category $\mathcal{C}$. We get a term of $\text{hom}_P(e, e')$ by noting we have functors $\square$

5 Operads

Definition 5.1 A family of operators is given by

1. $\mathcal{O}^\otimes : \mathcal{F}_* \rightarrow \mathcal{U}$
- 2.

$$\prod_{XY : \mathcal{F}_*} \prod_{f : X \rightarrow_{\mathcal{F}_*} Y} \text{inert}(f) \rightarrow \prod_{a : \mathcal{O}^\otimes(X)} \sum_{b : \mathcal{O}^\otimes(Y)} \sum_{g : a \rightarrow_{\mathcal{O}^\otimes(f)} b} \text{isCoCart}g$$

That is all inert arrows have cocartesian lifts

3. Let $a : \mathcal{O}^\otimes(X)$ and $b : \mathcal{O}^\otimes(Y)$. We may obtain for each $\rho : X \rightarrow \langle 1 \rangle$, a term $c : \mathcal{O}^\otimes(\langle 1 \rangle)$ and a cocartesian lift $f : b \rightarrow c_\rho$ over ρ . We ask that the natural map

6 A list of questions for Ulrik

Some modal type theory Let $C : \mathfrak{b} \mathcal{U}$ be a category. and $\phi : \# \langle \mathfrak{b} \mid C \rangle \rightarrow \text{Prop}_{\square}$. Define

$$C_\phi = \sum_{c : C} \langle \# \mid \phi(c^\eta) \rangle$$

How can we apply $\phi(c^\eta)$. Here we have $c : C$ so that $c^\eta : C^\eta$, but that's it. Can we get from C^η to $\langle flat \mid C \rangle$?